

Lezione 7 ricezione segnali polari binari

giovedì 16 aprile 2026 13:29

Riassumendo:

$\pm s(t)$
 $\pm s(t)$
 $\frac{m_0}{2}$
 $f_s = 10^6$ Hz
 $\omega = 2\pi f_s$
 $R(f) = S^*(f)$
 $f_b = f_s$
 $\sum_{N=1}^N \max s_i$ ha periodo
 $R(f) = S^*$
 $E = \int |s(t)|^2 dt \Rightarrow E_b = E$; $\sigma^2 = \frac{m_0}{2} E$
 $P_b = \frac{2}{3\pi} e^{-\frac{E_b}{\sigma^2}} \rightarrow \gamma$ $P_b = 10^{-6}$ $\gamma = 10,5$ dB

SEGNALI QUASITERNARI (a periodo di f_b la banda è la metà e a periodo di banda la f_b è doppia)

$\frac{E}{N} \rightarrow \frac{E_b}{m_0}$
 $\downarrow$ banda ed $\uparrow$ E_b/m_0
 $\uparrow$ bit rate ed $\uparrow$ $\frac{E_b}{m_0}$ (periodo di banda)

$\pm s(t)$
 $\pm s(t)$
 $\frac{m_0}{2}$
 $S^*(f)$
 ω
 $P_b = \frac{2}{3\pi} e^{-\frac{E_b}{\sigma^2}}$
 $\frac{E_b}{\sigma^2} = 9$ dB
 $\frac{E_b}{2,5}$
 $\pm s(t)$ E
 $\pm s(t)$ $4E$
 $E_{media} = 5E$
 $\frac{E_b}{2,5} = E$

SEGNALI OTTONARI

$\pm s(t)$
 $\pm s(t)$
 $\pm s(t)$
 $\pm s(t)$
 $\frac{m_0}{2}$
 $S^*(f)$
 ω
 $P_b = \frac{2}{9} e^{-\frac{E_b}{\sigma^2}}$
 $\frac{E_b}{\sigma^2} = 9,5$ dB
 $E = 4E + 25E + 69E = 94E_m$
 $E_b = 7E$

SEGNALI A 16 bit

$\pm s(t), \pm s(t), \pm s(t), \pm s(t),$
 $\pm s(t), \pm s(t), \pm s(t), \pm s(t),$
 $\pm s(t), \pm s(t), \pm s(t), \pm s(t)$
 $P_b = \frac{2}{9} e^{-\frac{E_b}{\sigma^2}}$
 $11,2 = 13$ dB

SEGNALI A 32 bit (simile a 16)
 $P_b = \frac{2}{9} e^{-\frac{E_b}{\sigma^2}}$
 $12,5 = 13$ dB

Riassunto Endo

binario $\frac{fb}{2}$ $\frac{e^{-\frac{\gamma}{2n}}}{3n}$
 quaternario $\frac{fb}{4}$ $\frac{1.5}{2} \frac{e^{-\frac{\gamma}{2n}}}{3n}$
 ottario $\frac{fb}{8}$ $\frac{7}{4} \frac{e^{-\frac{\gamma}{2n}}}{3 \cdot 3n}$
 16 livelli $\frac{fb}{8}$ $\frac{2}{4} \frac{e^{-\frac{\gamma}{2n}}}{6 \cdot 3n}$
 32 livelli $\frac{fb}{16}$ $\frac{e^{-\frac{\gamma}{2n}}}{5 \cdot 3n}$

Vediamo ora quelli **modulati** :

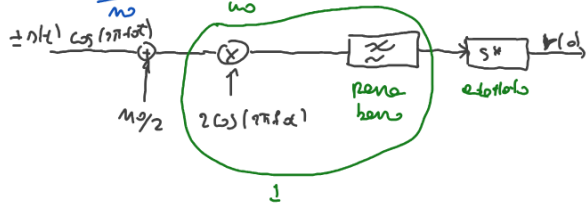
2PSK: $x(t) = \pm a(t) \cos(2\pi f_d t)$ → Modulation

$$\Downarrow$$

$$-r(t) \cos(2\pi f_{\text{osc}}) = r(t) \cos(2\pi f_{\text{osc}} + \pi)$$

$$P_b \cdot \frac{e^{-\delta}}{n}$$

$$\gamma_2 = \frac{E_b}{\omega_0}$$

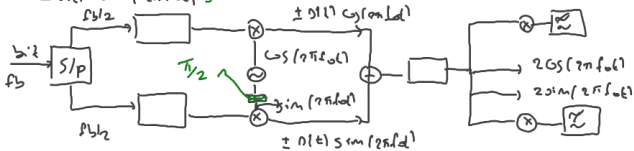


trasse 2. Linie von Alter mülle

Bremin: fb ; Sele: $L_5(n, p)$ $P_5 = \frac{e^{-\gamma}}{3\pi}$ $\gamma = \frac{E_5}{w_0}$

QPSK: 4 combinazioni / portanti

$$\left. \begin{aligned} \pm 5(t) \cos(2\pi f_0 t) \\ \pm 5(t) \sin(2\pi f_0 t) \end{aligned} \right\} \textcircled{\perp}$$



(2 sistemi PSK interrelazionati ed indipendenti)

$$B_{\min} = \frac{f_b}{2}; \quad B_{\text{eff}} = \frac{f_b}{2}(1+\rho); \quad P_b = \frac{e^{-\gamma}}{3\pi} \quad \gamma = \frac{E_b}{N_0}$$

1G RAM (asynchronous amplitude modulation)

sim
 ω
 $B_{\text{lin}} = \frac{f_b}{\omega} ; B_{\text{eff}} = \frac{f_b}{\omega} (1+p)$

6G QAM 3 livelli del seno; 3 livelli coseno

$$B_{min} = \frac{f_b}{c}; \text{ Refl} = \frac{f_b}{c} (1+p); P_b = \frac{7}{9} \frac{e^{-\frac{\gamma}{2}}}{33\pi}$$

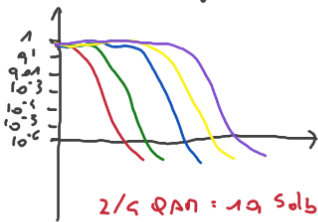
256 QPn 16 livelli del raso; 16 livelli coseno

$$B_{min} = \frac{f_b}{2}; B_{eff} = \frac{f_b}{2}(1+p); P_b = \frac{2}{\pi} \frac{e^{-1/2\pi}}{3\pi}$$

1026 QAN 32 livelli del raso; 32 livelli corono

$$B_{\text{min}} = \frac{f_b}{10}; B_{\text{eff}} = \frac{f_b}{10}(1+p); B_b = \frac{2}{3} \frac{e^{-\frac{\delta}{12.5}}}{3\pi}$$

Valores são graficamente:



2/5 QAM: 19,5 dB

16 QAM: 16,5 dB $\rightarrow$ 16 PSK $\rightarrow$ 16 dB

64 QAM: 19 dB $\rightarrow$ 64 PSK $\rightarrow$ 19,5 dB

256 QAM: 23,5 dB

1024 QAM: 29,5 dB

mo:

$$\frac{E_b}{\omega} \cdot \frac{1}{N} \leq \frac{S}{N} = \frac{E_b}{f_b} \cdot \frac{1}{N} \Rightarrow \frac{E_b}{f_b} \cdot \frac{1}{N} = \frac{E_b}{N \cdot f_b} = \frac{E_b}{N \cdot \frac{f_b}{h}}$$

(h = # bits)

$$\boxed{\frac{S}{N} = \frac{E_b \cdot h}{\omega_0}}$$